Fiscal Year:	FY 2010	**Task Last Updated:**	FY 04/09/2010
PI Name:	Saidel, Gerald Ph.D.		
Project Title:	Time Course of Metabolic Adaptations During Loading and Unloading		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:	NIH/National Institute of Biomedical Imaging and Bioengineering	**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSF 04-607 Interagency Solicitation Multi-Scale Modeling
Start Date:	03/01/2009	**End Date:**	12/31/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project actually ended 12/31/2009 per PI (Ed., 2/24/2012) NOTE: Project received extension until 3/31/2010 (from 3/31/2009) due to new PI Saidel in February 2009, per J. Dardano/JSC (3/2009)		
Key Personnel Changes/Previous PI:	NOTE: Gerald Saidel replaced Marco Cabrera as PI as of March 2009 (3/09 per J. Dardano/JSC). See Cabrera for reports from FY2006-FY2008.		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	The alterations in skeletal muscle structure and function induced by space travel or endurance training are the result of a chronic mechanical stimulus (or lack of it) a series of metabolic interactions spanning from the cellular to the organism level. Reduced activation of the weight bearing muscles alters gene expression of myosin heavy chain isoforms to meet the functional demands of weightlessness and results in muscle atrophy, reduced capacity to process fatty acids, and reduced muscle endurance. On the other hand, endurance training induces an up-regulation of oxidative enzymes and results in increased fatty acid oxidation capacity and muscle endurance. The few human and animal studies comparing skeletal muscle structure and function before and after unloading suggest that the adaptations are time and biological-level dependent. While some parameters may be altered within a week (muscle mass, insulin resistance), others may take longer to show changes (work efficiency, VO2max). Similar conclusions may be drawn from endurance training studies. Thus, to establish countermeasures aimed at restoring astronauts' work capacity, it is critical to determine: (1) the time course of muscle atrophy, (2) the bioenergetic, metabolic, and fuel-processing factors that contribute to the reductions in endurance associated with muscle atrophy, and (3) the effects of skeletal muscle atrophy on whole-body metabolism. To answer these questions a multi-scale model of energy metabolism -linking cellular to whole organism processes- is needed in combination with appropriate cellular to whole-body measurements collected during the period of adaptation. The objective of this proposal is to identify the time course of the adaptations to loading and unloading and to integrate them using a multiscale model of skeletal muscle and whole body metabolism, in order to predict the system integrated response after periods of space travel or endurance training. Accordingly, (1) A team of two computational modelers with expertise on scales ranging from cellular to whole body and 3 experimentalists with expertise in complementary methodologies in rodents (cell biochemistry [MAP], tracers and tissue metabolism [SFP], cardiorespiratory control [FTS] and whole body physiology [FTS, SFP]) has been formed to reach the proposal's objective. (2) A novel top-down approach will be applied to model whole-body metabolism and to bridge organism to cellular level. Thus, a minimal, but comprehensive, systems model of metabolism will be developed which includes: (a) brain, heart, liver, adipose tissue, and skeletal muscle, (b) a cardiorespiratory controller, (c) pancreas and liver, and (d) the cardiovascular system. Modeling transport processes across membranes will constitute an essential feature of the integration process linking the various scales. For model validation, molecular and cellular measurements will be used to assessed tissue and whole body metabolism. To maintain intact regulatory mechanisms, a novel methodological approach with emphasis on the use of in vivo measurements will be applied, specifically, proteomics and metabolomics using state-of-the-art stable isotope techniques. (3) This multi-scale model of metabolism will enable testing of physiologically-feasible time profiles of the adaptations to loading/unloading in every organ/tissue, which would otherwise require collection of hundreds of samples in thousands of animals. Specifically, this model can be used to predict tissue metabolism (e.g., fuel efficiency, buffering capacity, and muscle endurance) as well as whole body metabolism (e.g., insulin resistance, lactate acidosis, work capacity). The long-term goal of this project is to predict astronaut's energy metabolism and performance during altered loading states using a multi-level model of the human metabolic and energetic system in order to develop countermeasures. We envision that this computational model of human bioenergetics would become an essential component of the "Digital Astronaut". Thus, the specific aims of this proposal are: 1) To identify and model the time course of the structural, metabolic and functional adaptations to loading and unloading. 2) To develop and validate a multi-level (cell-muscle-whole body) computational model skeletal muscle metabolism in the context of the whole body. 3) To predict the integrated response of muscle fibers, whole skeletal muscle, and whole body during periods of loading and unloading, in particular astronaut's muscle energetics and function, as well as his/her whole body metabolism and performance. Involved with the Associate Director of NIH Center for Modeling Integrated Metabolic Systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Using a multiscale model of skeletal muscle and whole body metabolism will enable prediction of the system integrated response after periods of space travel or endurance training.
Task Progress:	Accomplishments In this research, we documented and validated models provide the desired simulations for integration into the Digital Astronaut program. In the model validation process, we compared simulations with experimental data from finished/published animal/human studies. The quantified validation of these models provides the Digital Astronaut project with defensible and credible evidence supporting the use of model simulations to predict muscle performance and atrophy. Model M.1: Simulations were compared with experimental data to validate model predictions of a) cellular metabolic responses from muscle atrophy (by unloading) and training (by exercise loading) of rodents b) gender differences and fiber types on oxygen uptake kinetics during exercise. Model M.2: Simulations were compared with oxygenation data from skeletal muscle measurements of venous blood and muscle tissue data with various levels of arterial oxygen content and exercise. This validation procedure ensured model robustness under hypoxic stress so that whole-body metabolic function can be evaluated. Model M.3: Simulations were compared with experimental data from skeletal muscle under ramp and constant load tests at different exercise intensities. This validation procedure identifed model limitations in predicting metabolic responses to higher exercise intensities. Model M.4: Simulations were compared with experimental data from cardio-respiratory and skeletal muscle responses to exercise in adults: (a) before and after training (b) before and after bed rest. We provided the Digital Astronaut program with uniform and well-documented computer codes that were or can readily be converted to another format, e.g., MATLAB. The documentation includes

	1. Background theory, variable, derivations, conceptual model structure, fundamental assumptions and implementation documentation including developers notes 2. Well-documented and commented code (FORTRAN and/or MATLAB) with references to all supporting documentation, including definition of all variables and units 3. Users guides that explain the implementation of the stand alone code 4. Outcome of all verification and validation testing, including determination of model limitations. 5. Copies of all publications resulting from this work
Bibliography Type:	Description: (Last Updated: 12/06/2010)
Articles in Peer-reviewed Journals	Li Y, Solomon TP, Haus JM, Saidel GM, Cabrera ME, Kirwan JP. "Computational model of cellular metabolic dynamics: Effect of insulin on glucose disposal in human skeletal muscle." Am J Physiol Endocrinol Metab. 2010 Jun;298(6):E1198-209. Epub 2010 Mar 23. PMID: 20332360 , Jun-2010